Cosmology after the Likelihood

Benjamin D. Wandelt

Anomaly detection

The Planck search for non-trivial topology

Benjamin Wandelt

How times change: **early 2000s**

Andrew approximates likelihoods

Ben MC samples hierarchical models

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RADICAL COMPRESSION OF COSMIC MICROWAVE BACKGROUND DATA

J. R. BOND[1]
Canadian Institute for Theoretical Astrophysics, Toronto, ON M5S 3H8, Canada
A. H. JAFFE[2]
Center for Particle Astrophysics, 301 LeConte Hall, University of California, Berkeley, CA 94720
AND
L. KNOX[3]
Canadian Institute for Theoretical Astrophysics, Toronto, ON M5S 3H8, Canada
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ABSTRACT

Powerful constraints on theories can already be inferred from *existing* CMB anisotropy data. But performing an exact analysis of available data is a complicated task and may become prohibitively so for upcoming experiments with $\gtrsim 10^4$ pixels. We present a method for approximating the likelihood that takes power spectrum constraints, e.g., "band-powers," as inputs. We identify a bias which results if one approximates the probability distribution of the band-power errors as Gaussian—as is the usual practice. This bias can be eliminated by using specific approximations to the non-Gaussian form for the distribution specified by three parameters (the maximum likelihood or mode, curvature or variance, and a third quantity). We advocate the calculation of this third quantity by experimenters, to be presented along with the maximum-likelihood band-power and variance. We use this non-Gaussian form to estimate the power spectrum of the CMB in 11 bands from multipole moment $\ell = 2$ (the quadrupole) to $\ell = 3000$ from all published band-power data. We investigate the robustness of our power spectrum estimate to changes in these approximations as well as to selective editing of the data.

Subject headings: cosmic microwave background — cosmology: miscellaneous — methods: data analysis — methods: statistical

PHYSICAL REVIEW D, VOLUME 70, 083511

Global, exact cosmic microwave background data analysis using Gibbs sampling

Benjamin D. Wandelt,[1,2,*] David L. Larson,[1] and Arun Lakshminarayanan[1]
[1]*Department of Physics, UIUC, 1110 W Green Street, Urbana, Illinois 61801, USA*
[2]*Department of Astronomy, UIUC, 1002 W Green Street, Urbana, Illinois 61801, USA*
(Received 16 October 2003; published 18 October 2004)

We describe an efficient and exact method that enables global Bayesian analysis of cosmic microwave background (CMB) data. The method reveals the joint posterior density (or likelihood for flat priors) of the power spectrum C_ℓ and the CMB signal. Foregrounds and instrumental parameters can be simultaneously inferred from the data. The method allows the specification of a wide range of foreground priors. We explicitly show how to propagate the non-Gaussian dependency structure of the C_ℓ posterior through to the posterior density of the parameters. If desired, the analysis can be coupled to theoretical (cosmological) priors and can yield the posterior density of cosmological parameter estimates directly from the time-ordered data. The method does not hinge on special assumptions about the survey geometry or noise properties, etc., It is based on a Monte Carlo approach and hence parallelizes trivially. No trace or determinant evaluations are necessary. The feasibility of this approach rests on the ability to solve the systems of linear equations which arise. These are of the same size and computational complexity as the map-making equations. We describe a preconditioned conjugate gradient technique that solves this problem and demonstrate in a numerical example that the computational time required for each Monte Carlo sample scales as $n_p^{3/2}$ with the number of pixels n_p. We use our method to analyze the data from the Differential Microwave Radiometer on the Cosmic Background Explorer and explore the non-Gaussian joint posterior density of the C_ℓ from the Differential Microwave Radiometer on the Cosmic Background Explorer in several projections.

DOI: 10.1103/PhysRevD.70.083511 PACS numbers: 98.80.-k, 02.50.Sk, 02.70.-c, 98.70.Vc

How times change: **now**

Andrew MC samples hierarchical models

KiDS+VIKING-450 cosmology with Bayesian hierarchical model redshift distributions

George T. Kyriacou,[1] Arrykrishna Mootoovaloo,[1,2] Alan F. Heavens,[1★] Andrew H. Jaffe,[1]
Florent Leclercq,[3] Konrad Kuijken[4]

[1] *Imperial Centre for Inference and Cosmology (ICIC), Imperial Astrophysics, Imperial College, Blackett Laboratory, Prince Consort Road, London, SW7 2AZ, UK*
[2] *Oxford Astrophysics, Department of Physics, University of Oxford, Denys Wilkinson Building, Keble Road, Oxford, OX1 3RH, UK*
[3] *CNRS & Sorbonne Université, UMR 7095, Institut d'Astrophysique de Paris, 98 bis boulevard Arago, F-75014 Paris, France*
[4] *Leiden Observatory, Leiden University, P.O. Box 9513, 2300 RA Leiden, The Netherlands*

Accepted XXX. Received YYY; in original form ZZZ

ABSTRACT

Tomographic redshift distributions from photometric data are crucial ingredients in cosmic shear analysis, since they are required for the theoretical calculation of the signal based on the redshift distribution of the galaxies where the shear field is sampled. In this paper, we develop as a proof of concept Leistedt et al.'s template-based Bayesian Hierarchical Model framework into an application to weak lensing data by sampling the redshift distributions of the galaxies in the KiDS+VIKING-450 survey. We also use a principal component analysis to provide a set of representative templates drawn from a large superset. For computational tractability, subsets of 10^5 galaxies are chosen to determine the redshift distributions, and we test the sensitivity of the cosmological inference to the subset chosen, finding it to be subdominant compared to the statistical error. We marginalise over the inferred redshift distributions and find that the Bayesian method increases the clustering parameter compared with previous studies, alleviating the S_8 tension with Planck, where $S_8 \equiv \sigma_8\sqrt{\Omega_{\rm m}/0.3} = 0.756 \pm 0.039$, assuming flat ΛCDM. The tension with Planck for this survey is reduced from 2.3σ to 1.9σ. We also infer a value for the matter density, $\Omega_{\rm m} = 0.31 \pm 0.10$.

Key words: gravitational lensing: weak – cosmology: cosmological parameters – techniques: photometric – methods: data analysis, statistical

Ben approximates likelihoods

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Advance Access publication 2018 March 28

Massive optimal data compression and density estimation for scalable, likelihood-free inference in cosmology

Justin Alsing,[1,2★] Benjamin Wandelt[1,3,4,5] and Stephen Feeney[1]

[1] *Center for Computational Astrophysics, Flatiron Institute, 162 5th Ave, New York City, NY 10010, USA*
[2] *Imperial Centre for Inference and Cosmology, Department of Physics, Imperial College London, Blackett Laboratory, Prince Consort Road, London SW7 2AZ, UK*
[3] *Institut d'Astrophysique de Paris (IAP), UMR 7095, CNRS UPMC Universite Paris 6, Sorbonne Universitse, 98bis boulevard Arago, F-75014 Paris, France*
[4] *Institut Lagrange de Paris (ILP), Sorbonne Universitse, 98bis boulevard Arago, F-75014 Paris, France*
[5] *Department of Physics and Astronomy, University of Illinois at Urbana-Champaign, 1002 W Green St, Urbana, IL 61801, USA*

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ABSTRACT

Many statistical models in cosmology can be simulated forwards but have intractable likelihood functions. Likelihood-free inference methods allow us to perform Bayesian inference from these models using only forward simulations, free from any likelihood assumptions or approximations. Likelihood-free inference generically involves simulating mock data and comparing to the observed data; this comparison in data space suffers from the curse of dimensionality and requires compression of the data to a small number of summary statistics to be tractable. In this paper, we use massive asymptotically optimal data compression to reduce the dimensionality of the data space to just one number per parameter, providing a natural and optimal framework for summary statistic choice for likelihood-free inference. Secondly, we present the first cosmological application of Density Estimation Likelihood-Free Inference (DELFI), which learns a parametrized model for joint distribution of data and parameters, yielding both the parameter posterior and the model evidence. This approach is conceptually simple, requires less tuning than traditional Approximate Bayesian Computation approaches to likelihood-free inference and can give high-fidelity posteriors from orders of magnitude fewer forward simulations. As an additional bonus, it enables parameter inference and Bayesian model comparison simultaneously. We demonstrate DELFI with massive data compression on an analysis of the joint light-curve analysis supernova data, as a simple validation case study. We show that high-fidelity posterior inference is possible for full-scale cosmological data analyses with as few as $\sim 10^4$ simulations, with substantial scope for further improvement, demonstrating the scalability of likelihood-free inference to large and complex cosmological data sets.

Key words: methods: data analysis.

The change in practice of cosmological statistics

Designing closed form likelihoods that strike a compromise between fidelity and inference tractability

Implicit likelihood defined by a generative procedure

A concept that bridges explicit and implicit inference

The Fisher score function

$$\nabla_\theta \log P(x|\theta)$$

Andrew was amongst the first to explore the Fisher score
in the context of cosmology!

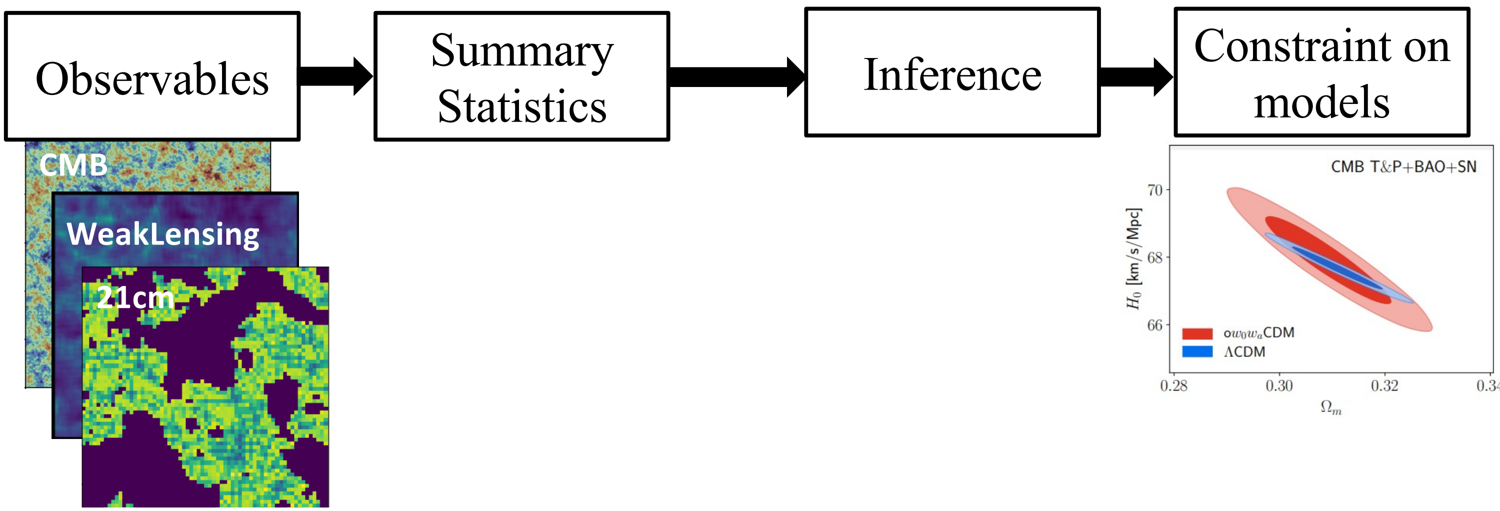
Fisher Score Matching
for
Simulation-Based Forecasting and Inference

Ce Sui, Shivam Pandey, and Benjamin D. Wandelt

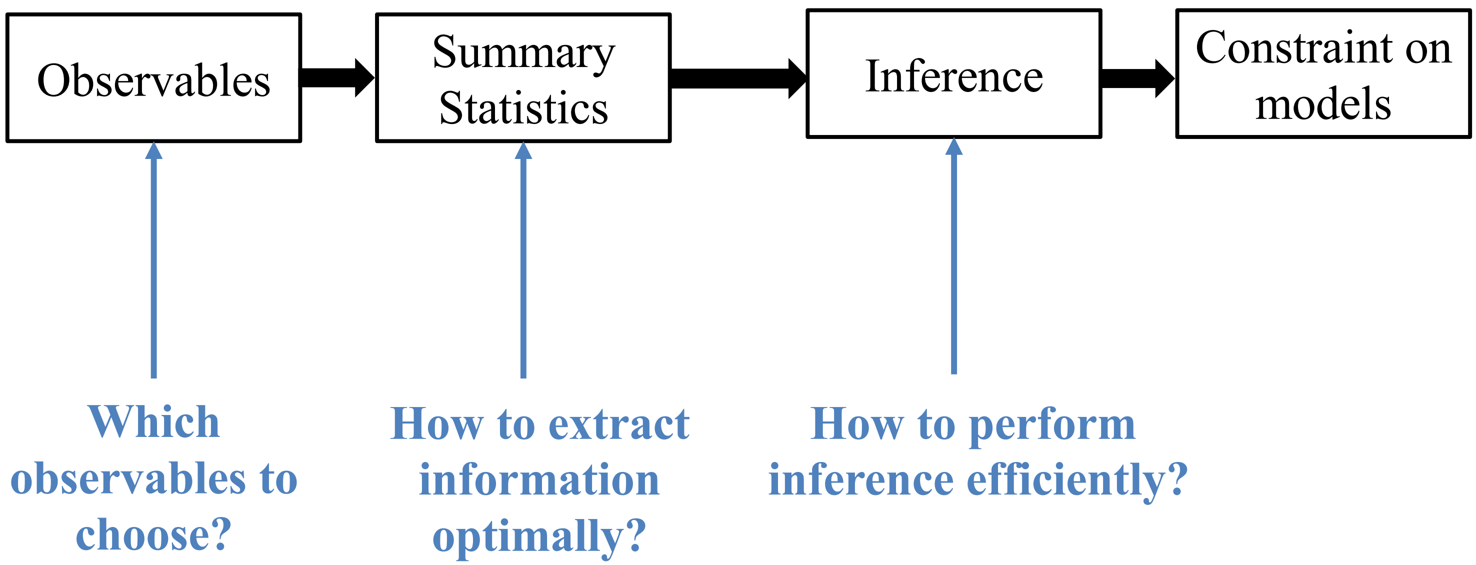
arXiv:2507.07833

ML4Astro, ICML2025

Statistical Inference in Cosmology



Statistical Inference in Cosmology



Fisher Score is All You Need

Given a physical model, the **Fisher Score** provides

$x|\theta \sim P(x|\theta)$ $\nabla_\theta \log P(x|\theta)$

- (Local) Information Content
- Sufficient Statistic
- Maximum Likelihood Estimation
- Gradient-Based Sampling

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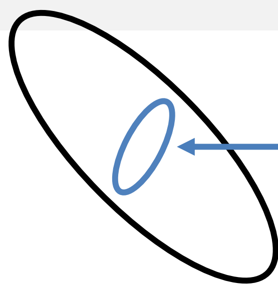
Fisher Information : $I(\theta) = E_{P(x|\theta)}[\nabla_\theta \log P(x|\theta)\, \nabla_\theta \log P(x|\theta)^T]$

Cramer-Rao Bound:
$Cov(\hat{\theta}) \succcurlyeq I(\theta)^{-1}$



CR-bound (Fisher information)

Any unbiased estimator $\hat{\boldsymbol{\theta}}$

Estimation Variance

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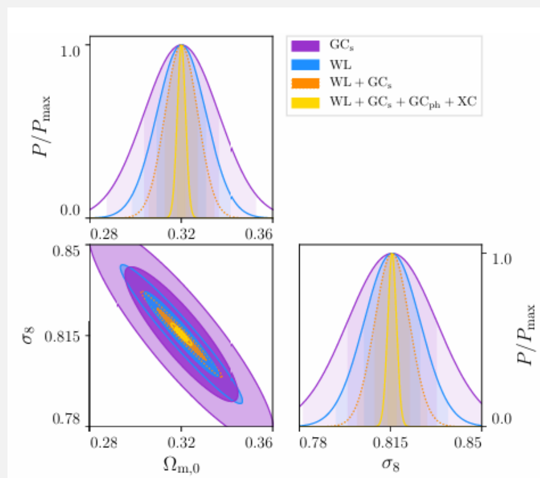
(Local) Information Content

Sufficient Statistic

Maximum Likelihood Estimation

Gradient-Based Sampling

Fisher Forecast



Example from Euclid preparation (arXiv:1910.09273, Fig 10)

Fisher Score is All You Need

Given a physical model, the **Fisher Score** provides

$x|\theta \sim P(x|\theta)$ $\nabla_\theta \log P(x|\theta)$

(Local) Information Content

Sufficient Statistic

Maximum Likelihood Estimation

Gradient-Based Sampling

Score Compression $s(x, \theta') = \nabla_{\theta'} \log P(x|\theta')$

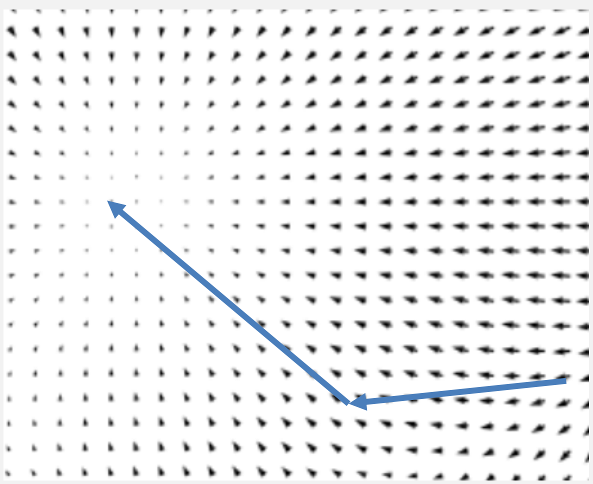
saturates the CR bound when θ' is near the true parameter.

Fisher Score is All You Need

Given a physical model, the **Fisher Score** provides

$x|\theta \sim P(x|\theta)$ $\nabla_\theta \log P(x|\theta)$

- (Local) Information Content
- Sufficient Statistic
- **Maximum Likelihood Estimation**
- Gradient-Based Sampling



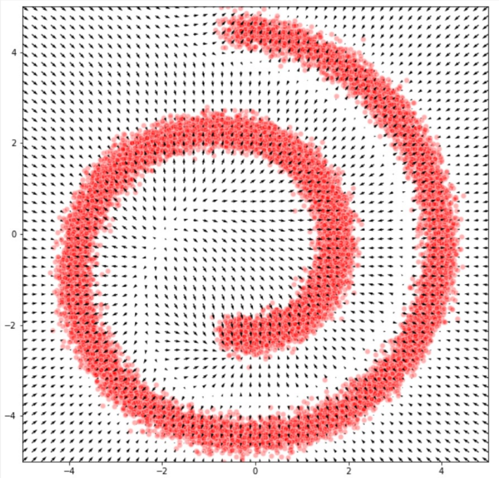
$$\theta_t = \theta_{t-1} + \lambda \nabla_\theta \log P(x|\theta)$$

Fisher Score is All You Need

Given a physical model, the **Fisher Score** provides

$x|\theta \sim P(x|\theta)$ $\nabla_\theta \log P(x|\theta)$

- (Local) Information Content
- Sufficient Statistic
- Maximum Likelihood Estimation
- **Gradient-Based Sampling**



$$\nabla_\theta \log P(\theta|x) = \nabla_\theta \log P(x|\theta) + \nabla_\theta \log P(\theta)$$

$$+$$

Gradient-based sampler

Limitations of current Fisher score applications:

1. **Difficult to estimate** — often requires finite differences or fully differentiable models.

2. **Gaussian likelihood assumption** — typically assumes the data likelihood is Gaussian.

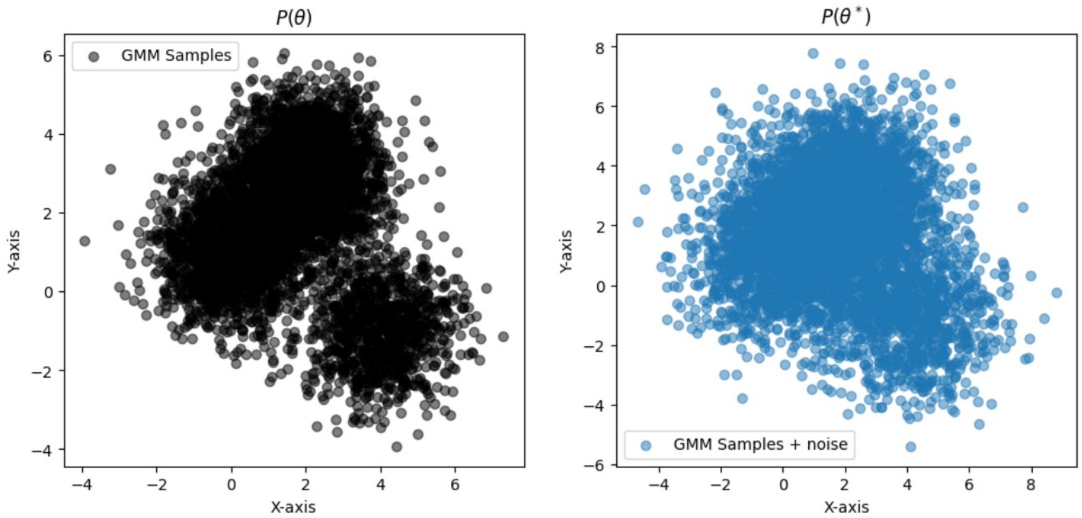
Score Matching in Diffusion Model

To sample from a target distribution $\boldsymbol{\theta} \sim \boldsymbol{p}(\boldsymbol{\theta})$, we can learn

$$\boxed{\nabla_\theta \log \boldsymbol{p}(\boldsymbol{\theta})}$$

We perturbed it to $\boldsymbol{\theta}^*$ by adding noise, i.e., $\boldsymbol{\theta}^* | \boldsymbol{\theta} \sim \boldsymbol{P}(\boldsymbol{\theta}^* | \boldsymbol{\theta})$:

$$\boxed{\boldsymbol{\theta} \rightarrow \boldsymbol{\theta}^*}$$

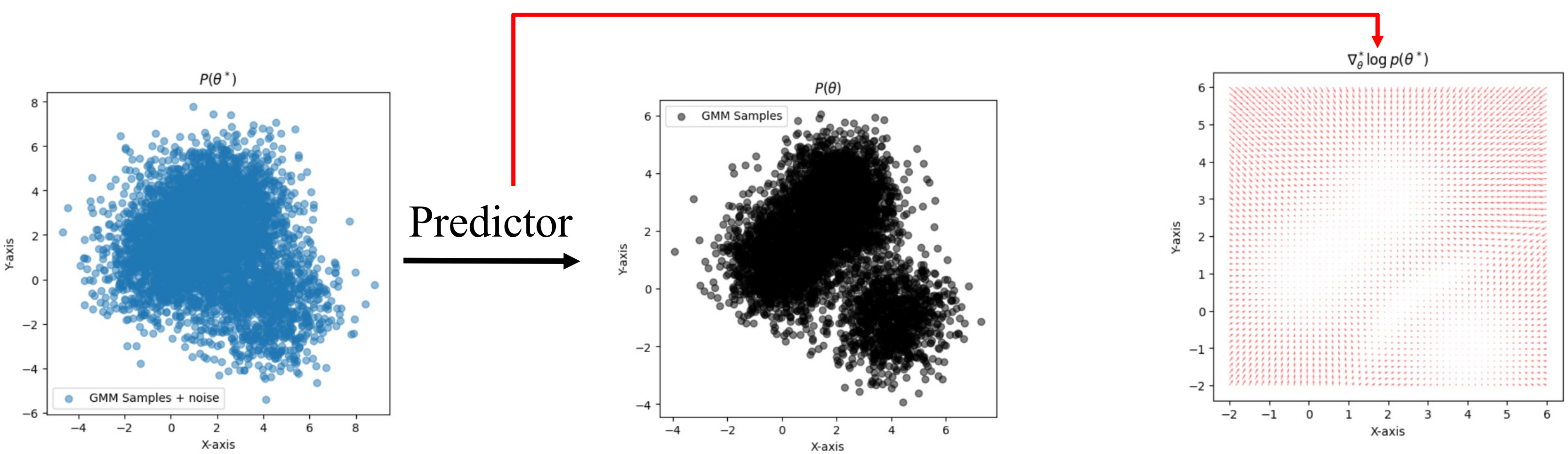


Score Matching in Diffusion Model

We can show that $$\nabla_{\theta^*} \log P(\theta^*) = E_{P(\theta|\theta^*)}[\nabla_{\theta^*} \log P(\theta^*|\theta)]$$

If $P(\theta^*|\theta)$ is Gaussian, then we are essentially learning a denoiser

$$\nabla_{\theta^*} \log P(\theta^*) = E_{P(\theta|\theta^*)}\left[\frac{\theta - \theta^*}{\sigma^2}\right] \Longleftrightarrow \sigma^2 \nabla_{\theta^*} \log P(\theta^*) + \theta^* = E_{P(\theta|\theta^*)}[\theta]$$



Estimate Fisher Score through Score Matching

Given a physical model $\boxed{\boldsymbol{\theta} \rightarrow \boldsymbol{x}}$

Assume the existence of an intermediate (latent) state $\boldsymbol{\theta}^*$ $\boxed{\boldsymbol{\theta} \rightarrow \boldsymbol{\theta}^* \rightarrow \boldsymbol{x}}$

We can show that

$$\nabla_{\boldsymbol{\theta}} \log P(\boldsymbol{x}|\boldsymbol{\theta}) = \boldsymbol{E}_{P(\boldsymbol{\theta}^*|\boldsymbol{x},\boldsymbol{\theta})}[\nabla_{\boldsymbol{\theta}} \log P(\boldsymbol{\theta}^*|\boldsymbol{\theta})]$$

Partial knowledge of the score yields the full score

Learning the Fisher score with neural networks

$\nabla_\theta \log P(x|\theta)$ **is the posterior mean of** $\nabla_\theta \log P(\theta^*|\theta)$

$$\nabla_\theta \log P(x|\theta) = E_{P(\theta^*|x,\theta)}[\nabla_\theta \log P(\theta^*|\theta)]$$

The posterior mean of a quantity can be estimated by training a neural network to predict it.

$$\nabla_\theta \log P(x|\theta) = \operatorname*{argmin}_s E_{p(x,\theta,\theta^*)}\left[(s(x,\theta) - \nabla_\theta \log P(\theta^*|\theta))^2\right]$$

If your latent model $\log P(\theta^*|\theta)$ **is Gaussian, then you are essentially predicting** θ^* **from** (x,θ)

Fisher Score Matching

$$\nabla_{\boldsymbol{\theta}} \log \boldsymbol{P}(\boldsymbol{x}|\boldsymbol{\theta}) = \underset{\boldsymbol{s}}{\operatorname{argmin}}\, \boldsymbol{E}_{\boldsymbol{p}(\boldsymbol{x},\boldsymbol{\theta},\boldsymbol{\theta}^*)}\left[\left(\boldsymbol{s}(\boldsymbol{x},\boldsymbol{\theta}) - \nabla_{\boldsymbol{\theta}} \log \boldsymbol{P}(\boldsymbol{\theta}^*|\boldsymbol{\theta})\right)^2\right]$$

1. **Sample from** $\boldsymbol{p}(\boldsymbol{x},\boldsymbol{\theta},\boldsymbol{\theta}^*)$, essentially run simulation $\boldsymbol{\theta} \rightarrow \boldsymbol{\theta}^* \rightarrow \boldsymbol{x}$
2. **Train network** $\boldsymbol{s}(\boldsymbol{x},\boldsymbol{\theta})$ **to predict** $\nabla_{\boldsymbol{\theta}} \log \boldsymbol{P}(\boldsymbol{\theta}^*|\boldsymbol{\theta})$
3. **Upon convergence,** $\boldsymbol{s}(\boldsymbol{x},\boldsymbol{\theta}) \approx \nabla_{\boldsymbol{\theta}} \log \boldsymbol{P}(\boldsymbol{x}|\boldsymbol{\theta})$

Decomposable and Indecomposable Models

Given a physical model

$$\boxed{\boldsymbol{\theta} \rightarrow \boldsymbol{x}}$$

Decomposable

Assume the existence of an intermediate (latent) state $\boldsymbol{\theta}^*$

$$\boxed{\boldsymbol{\theta} \rightarrow \boldsymbol{\theta}^* \rightarrow \boldsymbol{x}}$$

Indecomposable

Manually introduce a latent state $\widetilde{\boldsymbol{\theta}}$

$$\boxed{\widetilde{\boldsymbol{\theta}} \rightarrow \boldsymbol{\theta} \rightarrow \boldsymbol{x}}$$

and estimate $\nabla_{\widetilde{\boldsymbol{\theta}}} \log P\left(\boldsymbol{x} \middle| \widetilde{\boldsymbol{\theta}}\right)$

Applications of the Score Model

$$s^*(x, \theta) \approx \nabla_\theta \log P(x|\theta)$$

With this approach, we can:

1. Estimate the Fisher score for any simulation-based model (make non-differentiable model differentiable!)

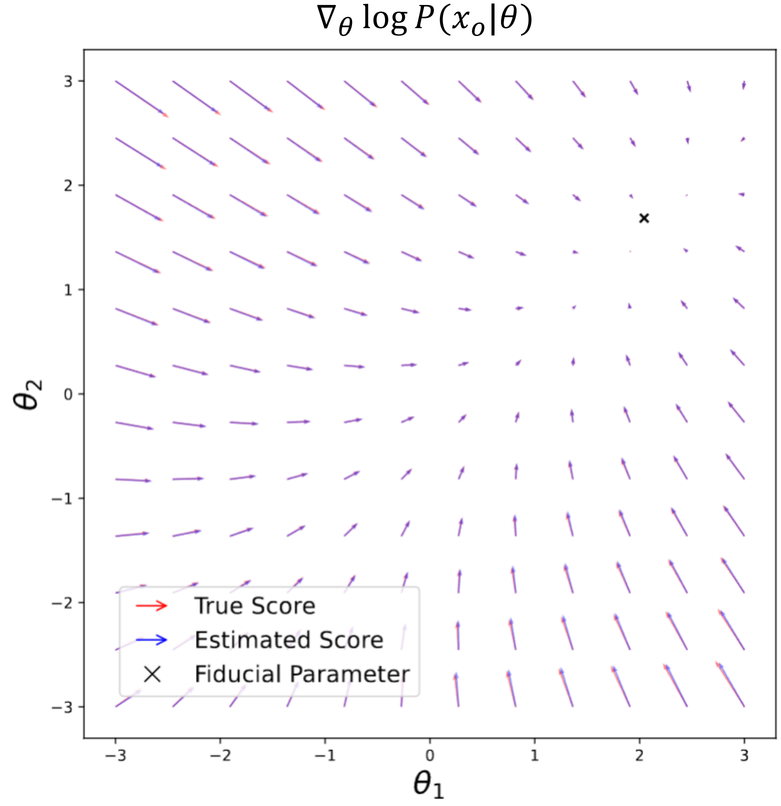
2. Perform Fisher forecasting without assuming Gaussianity, at any point in the trained parameter space

3. Carry out inference—both frequentist and Bayesian

Toy example: Linear Gaussian Model

$$x|\theta \sim N(\theta, \Sigma_n)$$

$$\theta^* \mid \theta \sim \mathcal{N}(\theta, \Sigma_{\theta^*}), \quad x \mid \theta^* \sim \mathcal{N}(\theta^*, \Sigma_n - \Sigma_{\theta^*})$$

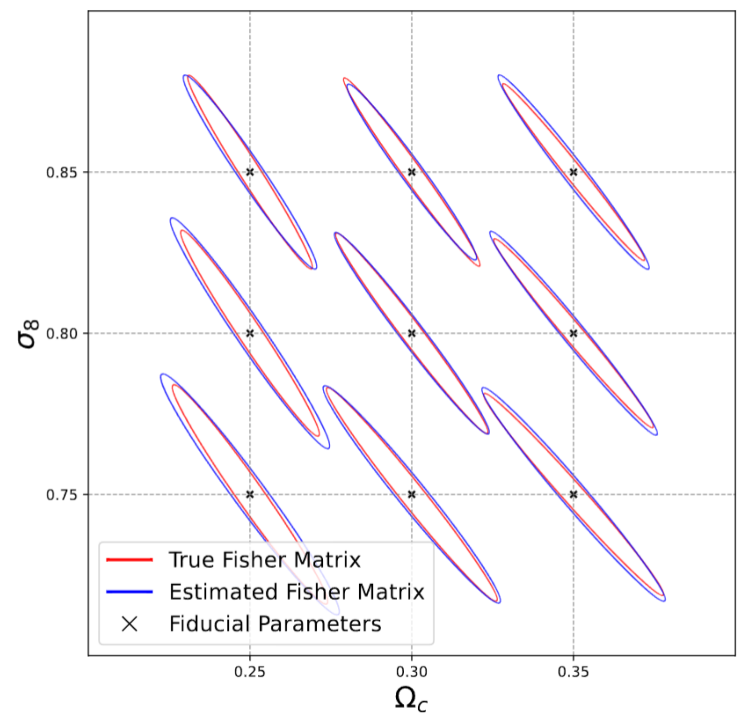


Cosmological Example: Weak Lensing Fisher Forecast

Parameters: Ω_c **and** σ_8

Observable: **Weak Lensing Angular Power Spectra**

Fisher Forecast compared to auto-differentiation



Jaxcosmo: https://github.com/DifferentiableUniverseInitiative/jax_cosmo

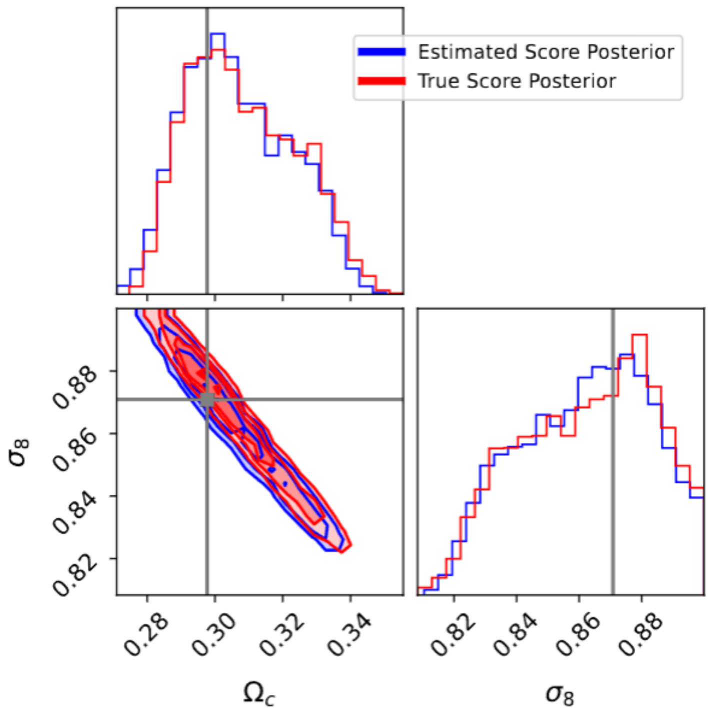
Cosmological Example: Weak Lensing Inference

Parameters: Ω_c **and** σ_8

Observable: **Weak Lensing Angular Power Spectra**

Sample posterior using HMC

$$\nabla_\theta \log P(\theta|x)$$
$$= \nabla_\theta \log P(x|\theta) + \nabla_\theta \log P(\theta)$$
$$\approx s^*(x,\theta) + \nabla_\theta \log P(\theta)$$



Jaxcosmo: https://github.com/DifferentiableUniverseInitiative/jax_cosmo

Summary

Given a physical model, the **Fisher Score** provides

$x|\theta \sim P(x|\theta)$ $\quad\quad$ $\nabla_\theta \log P(x|\theta)$

- Locally Optimal Statistic
- Fisher Forecast
- Maximum Likelihood Estimation
- Gradient-Based Sampling

We show that it can be estimated for **intractable likelihoods** and **non-differentiable models.**

Paper: https://arxiv.org/abs/2507.07833

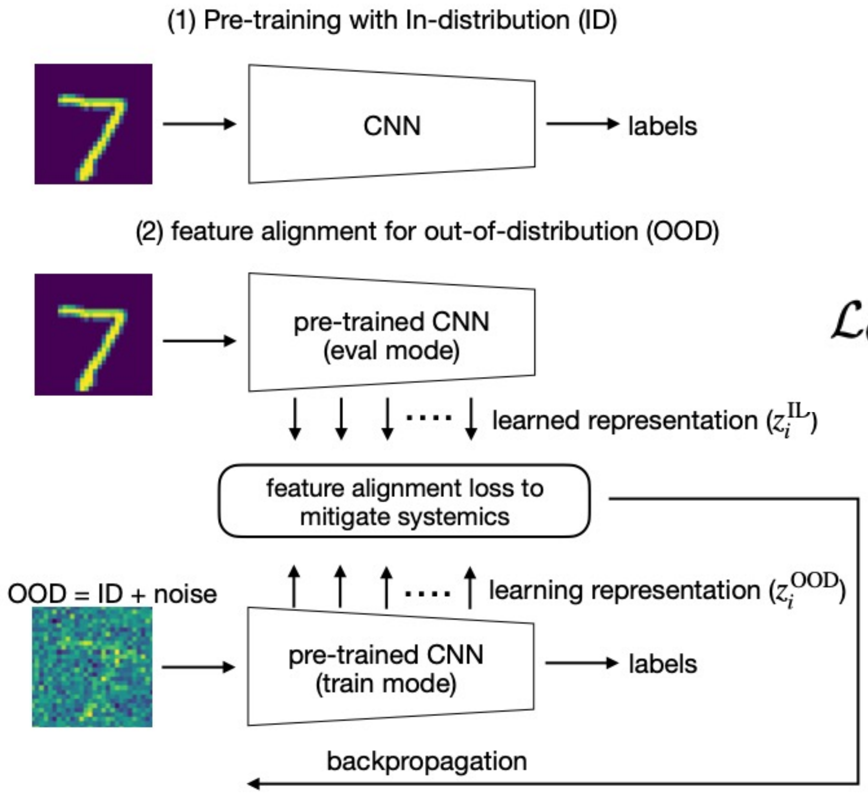
Codes: https://github.com/suicee/FisherScoreMatching

With more power comes great responsibility!

- Defining the model through a generative procedure gives a great deal more modeling freedom

- How to deal with model misspecification?

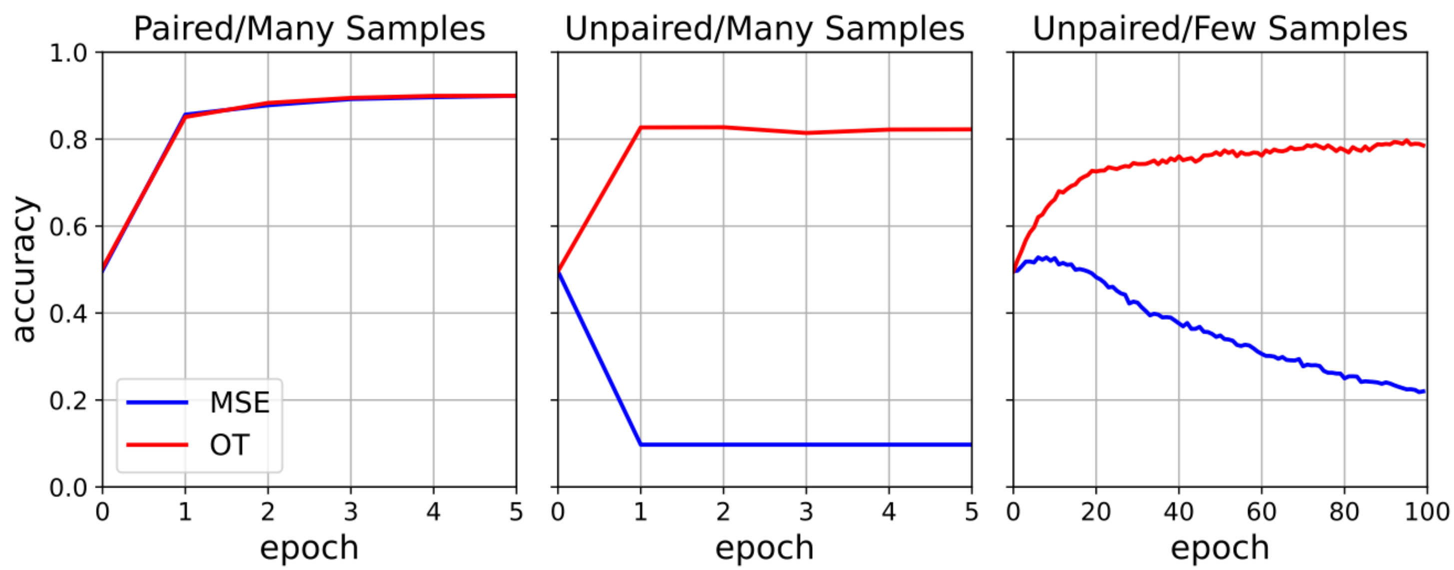
Feature Alignment with Optimal Transport



$$\mathcal{L}_{\mathrm{OT}} = \min_{\gamma \in \Pi(\mu,\nu)} \sum_{i,j} \gamma_{i,j}\, c(z_i^{\mathrm{ID}}, z_j^{\mathrm{OOD}})$$

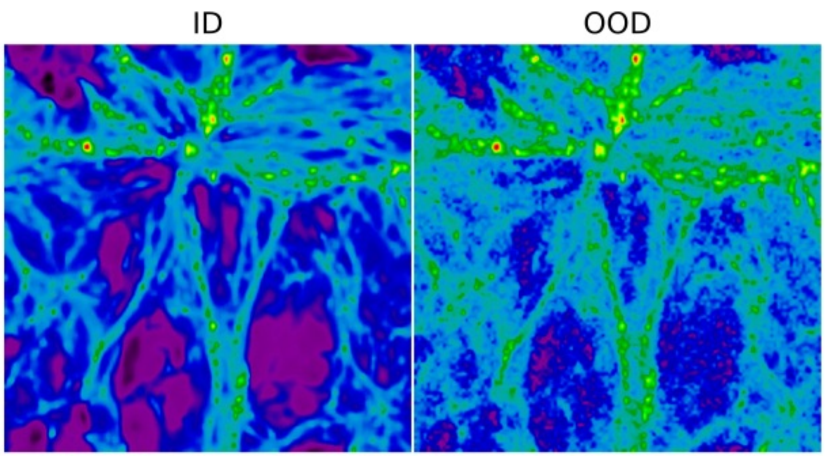
Hassan, Andrianomena, BDW arXiv:2511.11787

Feature Alignment with Optimal Transport



Hassan, Andrianomena, Wandelt arXiv:2511.11787

Feature Alignment with Optimal Transport



Alignment Scenario	R^2	Optimization steps
Paired/Many Samples	-5.3 → 0.86	1
Unpaired/Many Samples	-5.3 → 0.82	1
Unpaired/Few Samples	-5.3 → 0.86	50

Hassan, Andrianomena, Wandelt arXiv:2511.11787

Conclusion

- New methods have transformed the practice of machine learning
- Old concepts (Fisher score, OT) have renewed lives in this new age
- Andrew has been updating our beliefs for 60 years